Predictive Analytics for Taxi Services: A Case Study Using NYC TLC Data

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***Abstract*—This study presents a comprehensive analysis of New York City Taxi and Limousine Commission (TLC) trip data, focusing on fare patterns, tipping behavior, and predictive modeling. Using a subset of approximately $23,000$ rides, the project integrates data preprocessing, exploratory data analysis (EDA), hypothesis testing, regression modeling, and machine learning classification. Anomalies such as negative fares and flat-rate airport trips were identified and addressed. Hypothesis testing revealed that credit card payments are associated with significantly higher fares compared to cash. Linear regression models achieved an R^2 of 0.87 for fare prediction, while a random forest classifier accurately predicted generous tipping behavior ($\geq 20\%$) with an F1 score of 0.72. These findings demonstrate the utility of predictive analytics in urban transportation systems and provide actionable insights for drivers and policymakers. Future work will incorporate the full TLC dataset, contextual features, and explainable AI techniques to enhance fairness and transparency.**

***Index Terms*—Taxi fares, tipping behavior, New York City TLC, exploratory data analysis, regression, classification, machine learning.**

I. Introduction

Urban mobility systems generate massive amounts of data, providing opportunities to optimize transportation operations, pricing, and customer experience. The New York City Taxi and Limousine Commission (TLC) maintains detailed trip-level datasets that record fare amounts, trip distances, payment types, and tipping behavior. Analyzing this data can support drivers' revenue management, identify anomalous trips, and inform policy decisions related to urban transportation equity. However, TLC datasets often contain anomalies such as negative fares, extreme trip distances, and flat-rate airport fares, necessitating careful preprocessing. Predictive modeling can further aid in anticipating fare outcomes and tipping behavior, supporting both operational efficiency and customer satisfaction.

This study examines a representative TLC dataset ($23,000$ rides) to achieve three primary objectives:

- Conduct exploratory data analysis (EDA) to identify patterns and anomalies.
- Test hypotheses regarding payment methods and fare amounts.
- Develop predictive models for fare estimation and tipping classification.

Key contributions of this work include:

- A reproducible pipeline for data cleaning, anomaly detection, and feature engineering.
- Statistical evidence linking payment type to fare amount differences.
- Regression and machine learning models that provide actionable insights for fare prediction and tipping behavior.

This work is structured as follows: Section II describes the dataset and methodology; Section III presents EDA results; Section IV details modeling results; Section V discusses limitations; Section VI concludes with future directions.

II. Data & Methodology

A. Data Sources

The TLC dataset used in this study contains trip-level records for New York City taxi rides, including the following fields:

timestamps of trip start and end:
$pickup\_datetime$, $dropoff\_datetime$
distance traveled in miles: $trip\_distance$
fare components: $fare\_amount, tip\_amount, total\_amount$
payment-type:
1:$creditcard$, 2:$cash$, 3:$nocharge$, 4:$dispute$, 5:$unknown$.
categorical identifiers:
$RatecodeID, VendorID, pickup\_location\_id$,
$dropoff\_location\_id$
The dataset contains $23,000$ rides and represents a subset of the TLC records.

B. Data Preprocessing

Preprocessing included:

- Outlier detection and correction: Negative fares and durations were set to zero; extreme fares capped at \$62.50 based on interquartile range analysis.
- Feature engineering:
  - Created $mean\_distance$ and $mean\_duration$ for each pickup–dropoff pair.
  - Added temporal features: day_of_week, $month$, $rush\_hour$ indicator.
  - Engineered a binary generous target for tipping $\geq 20\%$.
- Encoding categorical variables: Converted categorical fields to dummy variables for modeling.

C. Tools and Libraries

The entire analysis was conducted using Python, primarily within a Jupyter Notebook environment. Key libraries included:

- $pandas, numpy$ – data wrangling and preprocessing
- $matplotlib, seaborn, plotly$ – static and interactive data visualization
- $scikit-learn$ – regression and classification models
- $xgboost$ – gradient boosting classifier

All code and results are available in a public GitHub repository for reproducibility.

D. Methodological Overview

The workflow consists of three main components:

- Exploratory Data Analysis (EDA): Identification of anomalies, distributions, and correlations.
- Hypothesis Testing: Comparison of fare amounts across payment types using t-tests.
- Predictive Modeling:
  - Linear regression for fare prediction.
  - Random forest and XGBoost classifiers for tipping behavior.

III. Exploratory Data Analysis

A. Data Overview

Initial inspection revealed no missing values but included anomalous fares (negative or extremely high) and extreme trip distances. Most trips ranged from 1–3 miles with fares typically below $30, except for flat-rate airport trips.

B. Payment Type Analysis

Average fare for credit card transactions was higher than for cash, suggesting a correlation between payment type and trip value.

C. Temporal Patterns

Rides were more frequent during weekday rush hours ($06:00$–$10:00, 16:00$–$20:00$). Mean fare also varied slightly by day of the week.

D. Correlation Analysis

$mean\_distance$ and $mean\_duration$ were highly correlated with $fare\_amount (Pearson r = 0.87)$, indicating strong predictive potential for regression models. The pairwise plot and the heatmap correlation matrix are shown in Fig 1, Fig 2

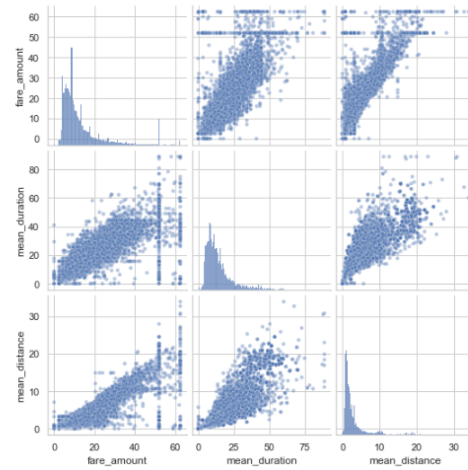

Fig. 1. Pairwise relationships between variables in the data

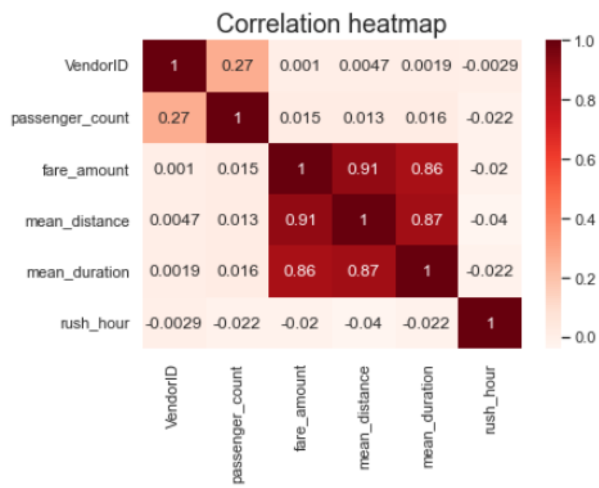

Fig. 2. Pearson pairwise correlation coefficient matrix

IV. Results & Interpretation

A. Hypothesis Testing on Payment Methods

To investigate whether payment type is associated with systematic differences in fare amounts, a $two-samplet-test$ was conducted comparing trips paid by credit card against those paid in cash. The null hypothesis assumed no difference in mean fares, while the alternative posited that a significant difference exists.

- $H0$: No difference in mean fare.
- Ha: Difference exists.

The test produced a highly significant result ($P < 10^{-11}$, leading to rejection of the null hypothesis.

The distribution of credit card fares exhibits a higher mean and greater variance compared to cash fares. One plausible explanation is that card payments are more common for longer trips, including airport routes with flat fares, whereas cash payments may dominate for shorter, intra-city rides. This finding aligns with prior research highlighting that passenger payment choice often correlates with trip length, convenience, and fare size.

B. Regression-Based Fare Prediction

A multiple linear regression model was developed to predict trip fares using distance, duration, and auxiliary features such as vendor identifiers. After standardization of predictors, the model demonstrated strong predictive power, with $R^2 = 0.87$ on the test set and a root mean squared error $(RMSE)$ of \$3.79. You see the plot of actual vs predicted values in Fig 3

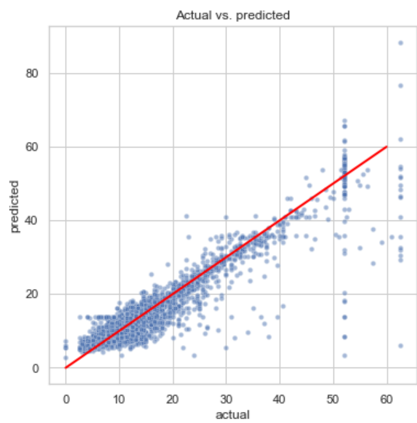

Fig. 3. Scatterplot of ‘predicted‘ over ‘actual

Inspection of model coefficients confirmed that trip distance was the most influential predictor, contributing approximately \$2 per mile in unstandardized units. The residual analysis Fig 4 indicated approximate normality and homoscedasticity, suggesting that the linear model adequately captured the main structure of the data. Importantly, while the model provides interpretable estimates, it does not account for unobserved factors such as traffic congestion or tolls, which may explain residual variance.

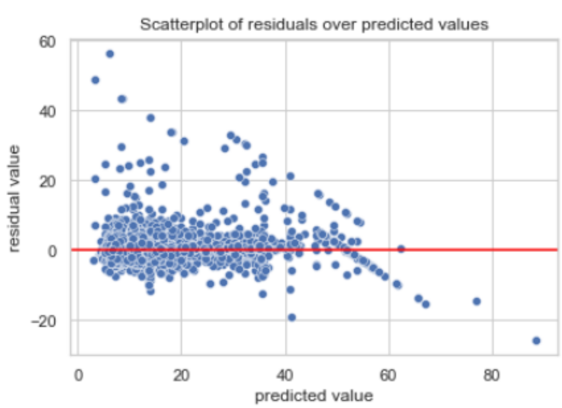

Fig. 4. Residuals over predicted values

C. Classification of Tipping Behavior

In addition to fare prediction, a classification task was conducted to predict whether passengers provided a “generous” tip, defined as 20% or more of the fare. $RandomForest$ and $XGBoost$ models were evaluated. The $RandomForest$ achieved an $accuracy$ of 68.6% and an $F1-score$ of 0.723, outperforming $XGBoost$ slightly (68.7% $accuracy$, $F1-score$ 0.710).
Feature importance analysis revealed that predicted fare, trip distance, and duration were strong determinants of tipping behavior. Vendor identifiers also contributed, suggesting possible differences in service patterns between providers Fig 5.

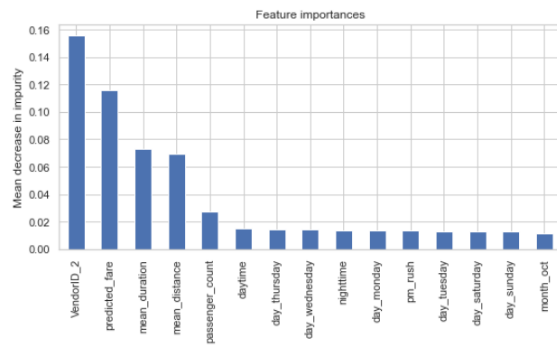

Fig. 5. Feature Importance Plot

Overall, these findings indicate that ensemble models can provide useful predictions of passenger tipping, though interpretability remains limited.

V. LIMITATIONS

Despite encouraging results, this study faces several limitations. First, the analysis is based on a limited sample of approximately $23,000$ trips, which may not fully capture the heterogeneity of the complete TLC dataset spanning millions of records. Second, several contextual factors known to influence taxi demand and fares—including weather, traffic conditions, and passenger demographics—were unavailable, potentially omitting important explanatory variables. Third, while ensemble models such as $RandomForests$ offer strong predictive $accuracy$, they reduce interpretability, limiting their value for causal inference or policy insights. Fourth, the dataset reflects trips from a single period; seasonal effects and long-term trends could alter patterns in tipping and fare structures. Finally, ethical considerations arise in tip prediction: deploying such models in real-world systems could inadvertently influence driver behavior, raising concerns around fairness and passenger privacy.

VI. CONCLUSION AND FUTURE WORK

This paper demonstrates the utility of predictive analytics for extracting actionable insights from urban transportation data. Through rigorous preprocessing, hypothesis testing, and predictive modeling, several contributions were achieved. First, anomalies in the dataset were identified and addressed, ensuring data quality for downstream tasks. Second, hypothesis testing confirmed that payment method is significantly associated with fare levels, with card transactions yielding higher average fares. Third, regression models successfully predicted fares with high accuracy, while classification models provided meaningful predictions of tipping behavior.

Looking forward, future work will pursue several directions. Expanding the analysis to the full TLC dataset will provide more robust generalization across time and geography. Incorporating contextual features—such as weather conditions, traffic density, and socio-demographic information—may substantially improve predictive performance. Finally, efforts will focus on explainable AI techniques to balance predictive accuracy with transparency, enabling fairer and more interpretable applications of machine learning in transportation analytics.

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